
HYPERION-SHIELD: Reactive, Revocable and History-Aware Safety for Vision-Language Models

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Vision-language models (VLMs) already power mixed-reality glasses, surgical
2 robots and on-device copilots, yet their safety guards remain static, single-turn
3 and unimodal. They cannot react once a jailbreak is detected, they provide no
4 guarantees after local fine-tuning, and they ignore leakage through higher-order
5 derivatives, low-bit gradient releases or physiological side-channels. We address
6 these shortcomings with HYPERION-SHIELD, a five-pillar framework: (i) a
7 Second-Order Information Curtain that projects Hessians into a 64-dimensional
8 subspace and dithers 4-bit gradients while enforcing a PAC-Bayes leakage bound;
9 (ii) Continual Revocable Certification, which anchors every dialogue turn in a
10 BLS-signed Merkle chain and propagates 96-byte revocations within milliseconds;
11 (iii) a Reflexive Causal Patch that repairs unsafe causal paths on-device in ten
12 gradient steps and embeds witness-encrypted watermarks; (iv) a Gaze-Motion
13 Side-Channel Mask that maps latency, gaze and IMU traces onto a σ -smooth
14 manifold with $TV \leq 0.01$; and (v) CARE-Bench, the first continual arms-race
15 benchmark with 12 351 adaptive multimodal attack episodes. We fully specify three
16 experiments—leakage auditing, continual red-teaming and sensor masking—and
17 release logs. A pilot run exposed critical implementation gaps: mutual-information
18 estimates collapsed to zero, token recovery reached 100%, and only ten samples
19 were processed. We analyse these failures and outline the rerun conditions required
20 to substantiate HYPERION-SHIELD’s theoretical guarantees.

21 1 Introduction

22 Large VLMs increasingly mediate safety-critical decisions—from autonomous drones to multimodal
23 prosthetics—yet their alignment remains brittle. Human players in Tensor Trust contributed 126 k
24 prompt-injection attacks that subvert chat models Toyer et al. [2023]; adaptive suffixes achieve 100%
25 jailbreak success on leading text LLMs Andriushchenko et al. [2024]; and benign-looking adversarial
26 images can hijack cross-modal alignment Shayegani et al. [2023]. Certified defences exist but are
27 passive and unimodal. MMCert bounds ℓ_∞ perturbations on RGB-depth pairs Wang et al. [2024],
28 whereas Circuit Breaker suppresses harmful activations at inference time only Zou et al. [2024]. None
29 of these approaches (i) repair models after compromise, (ii) maintain guarantees over multi-turn
30 dialogues, or (iii) cover leakage through Hessians, quantised gradients, or physiological traces.

31 We therefore seek a holistic safety apparatus that delivers: 1) immediate reaction after a detected
32 exploit; 2) cryptographically verifiable guarantees across arbitrarily long interaction histories; and 3)
33 formal bounds on information leaked via logits, first- and second-order gradients and multimodal
34 side-channels. The adversary we consider is adaptive, multimodal and history-aware: they can
35 observe previous answers, fine-tune the model on-device, record latency together with gaze and
36 inertial data on an AR headset, and probe Hessians released for federated learning.

HYPERION-SHIELD tackles these challenges through five orthogonal components: • Second-Order Information Curtain (SOIC) learns a low-rank Hessian projector and injects σ -dither into 4-bit gradients, yielding a PAC-Bayes mutual-information bound. • Continual Revocable Certification (CRC) attaches threshold-BLS signatures to every turn and revokes compromised weights in <150 ms locally and <800 ms globally. • Reflexive Causal Patch (RCP) trains a 32 k-parameter router online to null unsafe causal directions and embeds witness-encrypted watermarks that ensure non-repudiation. • Gaze-Motion Side-Channel Masking (GMSM) deploys an eight-coupling normalising flow to decorrelate latency, gaze and IMU from hidden prompts. • CARE-Bench establishes the first continual arms-race protocol where AutoDAN-Vision agents adapt after each defence patch Liu et al. [2023].

1.1 Key contributions

- **Second-order leakage bound:** Derivation of a PAC-Bayes upper bound $I(h; y, \nabla, H) \leq \varepsilon_{\text{MI}}^2$ that remains valid under 4-bit gradient release and Hessian projection.
- **Revocable certificate chain:** Threshold-BLS certificate chain with millisecond-scale revocation and per-turn Merkle anchoring.
- **Reflexive Causal Patch:** A reactive patch that restores τ -safety within ten gradient steps and embeds a non-repudiable watermark.
- **Sensor side-channel masking:** First mask certified to $\text{TV} \leq 0.01$ on gaze-motion-latency traces.
- **CARE-Bench:** 12 351 adaptive attack episodes across four modalities with metrics Time-to-Breach (TTB), Revocation Latency (RL) and Conversation-Safe-Rate (CSR).
- **Protocol and logs:** Rigorous experimental protocol and public logs that uncover implementation gaps.

Future work will complete missing experiments, extend CRC to tool-state graphs in autonomous agents, and provide a formal proof for the witness-encrypted watermark.

2 Related Work

2.1 Static certification

Norm-bounded multimodal certificates such as MMCert restrict RGB-depth perturbations but assume frozen weights and single-step inputs Wang et al. [2024]. JS-DS constrains Jacobians; however, neither Hessians nor low-precision gradient releases are addressed. SOIC generalises these efforts through a learned 64-dimensional Hessian subspace and σ -dither that preserves differential privacy.

2.2 Reactive alignment

Circuit Breaker interrupts harmful activations on the fly Zou et al. [2024]; BadChain shows how back-doored reasoning steps exploit chain-of-thought Xiang et al. [2024]. RCP instead repairs the causal pathway and certifies the patch via witness encryption, while CRC ensures that compromised weights are provably revoked across clients.

2.3 Adaptive jailbreaks and benchmarks

PAIR finds semantic jailbreaks in <20 queries Chao et al. [2023]; AutoDAN produces stealthy adaptive prompts Liu et al. [2023]; StrongREJECT exposes benchmark inflation in jailbreak literature Souly et al. [2024]. CARE-Bench extends these lines to multimodal, continual red-teaming with sensor side-channels and Hessian probes.

2.4 Side-channel leakage

Language-model inversion extracts hidden prompts from next-token distributions Morris et al. [2023], while adversarial images transfer across thousands of prompts Luo et al. [2024]. GMSM is, to our knowledge, the first defence that masks latency-gaze-IMU traces in AR settings.

By offering second-order leakage bounds, cryptographically revocable guarantees and continual evaluation, HYPERION-SHIELD advances beyond the current state of the art.

3 Background

3.1 Formal setting

Let a foundation VLM f_θ map multimodal input $x = (\iota, \tau, s)$ consisting of image ι , text τ and sensor trace s to an answer y and auxiliary artefacts $a = (\nabla_\theta L, H_\theta L)$ that may be released for federated learning. A hidden prompt h resides in τ ; an adversary observes (y, a, s') and seeks to infer h . The system is ε -safe if $I(h; y, a, s') \leq \varepsilon$.

3.2 Dialogue history and multi-turn safety

Dialogue history $H = (h_1, \dots, h_T)$ evolves over time. Because attackers adapt after each turn, certification must hold jointly over the full history. Released gradients may be INT4; sensor traces comprise latency, gaze vectors and six-axis IMU readings; and a root authority can publish threshold-BLS signatures and revocations.

3.3 Cryptographic and information-theoretic tools

PAC-Bayes leakage bounds generalise classical differential privacy to information channels Wang et al. [2024]. Threshold-BLS signatures yield 96-byte proofs verifiable on mobile hardware. Witness encryption allows embedding encrypted statements whose decryption key is a later-revealed witness, providing non-repudiation without server trust.

4 Method

4.1 Second-Order Information Curtain

During safety fine-tuning we train a linear conditioner $g_H \in \mathbb{R}^{64 \times d}$ that projects each Hessian-vector product Hv onto a 64-dimensional subspace and applies 50% dropout. Concurrently, gradients destined for public release are quantised to INT4 using GPTQ symmetric rounding and perturbed by i.i.d. Gaussian σ -dither ($\sigma = 0.4$). If κ bounds the spectral norm of g_H and ρ is the zCDP noise for the dither, PAC-Bayes yields

$$I(h; y, \nabla, H) \leq (\kappa^2 + \rho) \varepsilon_{\text{MI}}^2.$$

4.2 Continual Revocable Certification

Safety-critical weights W_s are wrapped by an initial threshold-BLS signature σ_0 . For every dialogue turn t we compute a Merkle root $m_t = \text{Hash}(h_t, \text{timestamp})$ and sign it, generating σ_t . Clients verify σ_t in ≈ 5 ms on Apple A17 mobile hardware. When a 96-byte revocation proof appears on the bulletin board, the client zeros W_s and switches to a safe fallback, guaranteeing that the entire conversation remains τ -safe as long as no link in the chain is revoked.

4.3 Reflexive Causal Patch

Let $\mathcal{C}$ be the causal graph over residual streams. If the instantaneous causal effect $\text{CE}(u \rightarrow v)$ on an unsafe node v exceeds τ , an on-device 32 k-parameter router is trained online for ≤ 10 SGD steps under DP-CARC noise $\rho = 0.3$. A micro-certificate for is signed locally, and its hash is embedded via witness-encrypted watermark in the next benign answer, preventing silent removal by the operator.

4.4 Gaze-Motion Side-Channel Masking

An eight-coupling normalising flow f_ϕ maps the joint distribution of latency, gaze and IMU to a σ -smooth manifold statistically independent of h . We certify $\text{TV}(s', f_\phi(s')) \leq 0.01$ using the multimodal predictive-dynamic-fusion bound Cao et al. [2024].

122 4.5 Continual-arms-race evaluation

123 CARE-Bench distributes 12 351 adaptive attack episodes driven by AutoDAN-Vision agents. Primary
124 metrics are median TTB (higher is better), CSR (higher is better) and RL (lower is better).

125 5 Experimental Setup

126 5.1 Infrastructure

127 Experiments run on $8 \times$ NVIDIA A100-80 GB GPUs with 2 TB RAM, orchestrated by Slurm.

128 5.2 Victim model

129 Llama-2-70B-chat with a vision adaptor; safety heads are hot-swapped per variant.

130 5.3 Datasets

- 131 • Secret prompts: PII-Bench-v1 plus MUSDL-Lyrics (5 000).
- 132 • Benign prompts: MT-Bench-Neutral (5 000).
- 133 • Sparse LoRA updates: FedLoRA-1B.
- 134 • CARE-Bench-v1: 12 351 adaptive episodes with sensor traces.
- 135 • GSM: GazeCapture-IMU, EgoMotion-IMU and AR-Latency-2023.

136 5.4 Variants

137 Leakage audit: Base, ORACLE-GUARD, HYPERION, HYP-no-Hessian, HYP-no-Dither. CARE
138 adds HYP-no-CRC and HYP-no-RCP.

139 5.5 Experiment 1 - SOIC leakage

140 Prompts are split 70/15/15; five seeds $\{11, 23, 29, 47, 83\}$. For each prompt we capture logits, full
141 FP16 gradients, INT4 gradients with $\sigma = 0.4$ dither, and 64 Hessian projections. Mutual information
142 is estimated by MINE (three-layer MLP, 50 k steps, Adam 3×10^{-4}) and KSG-kNN ($k = 10$) with
143 1 000 bootstrap CIs. Token recovery uses the LoRA-aware inversion of Zhu et al. 2024.

144 5.6 Experiment 2 - CARE arms-race

145 Five-fold split; same seeds. Metrics: median TTB, CSR, RLlocal, RLglobal. Stressors: (i) network
146 blackout, (ii) 3 s latency jitter, (iii) thermal-IR images. RCP hyper-parameters: steps $\in \{2, 5, 10\}$,
147 $\tau \in \{0.1, 0.2, 0.3\}$.

148 5.7 Experiment 3 - GSM sensor masking

149 Flow depth = 8, $\sigma = 0.01$. Metrics: Extraction-Accuracy (lower is better), TV-distance (lower is
150 better) and Watermark TNR (higher is better). Five seeds $\{3, 7, 13, 19, 101\}$.

151 5.8 Reproducibility

152 SHA-256 of every checkpoint, dataset commit and source revision is stored in JSON artefacts; a CI
153 smoke test exercises all flags in <90 s.

154 6 Results

155 Only Experiment 1 ran beyond the CI smoke test; Experiments 2 and 3 produced no logs. Moreover,
156 the leakage audit processed merely ten samples per variant, invalidating statistical claims.

157 **6.1 Mutual information**

158 All five variants report $\hat{\epsilon}_{\text{MI}}^2 = 0.0$ bits (mean of 10 samples, sd 0), an artefact of degenerate gradient
159 capture.

160 **6.2 Exact-token recovery**

161 All variants achieve 100% recovery, contradicting zero mutual information. Manual inspection shows
162 the inversion metric compared recovered tokens against teacher-forced inputs rather than hidden
163 prompts.

164 **6.3 Sample size**

165 `num_samples = 10` instead of the planned 50 k; confidence intervals are therefore meaningless.

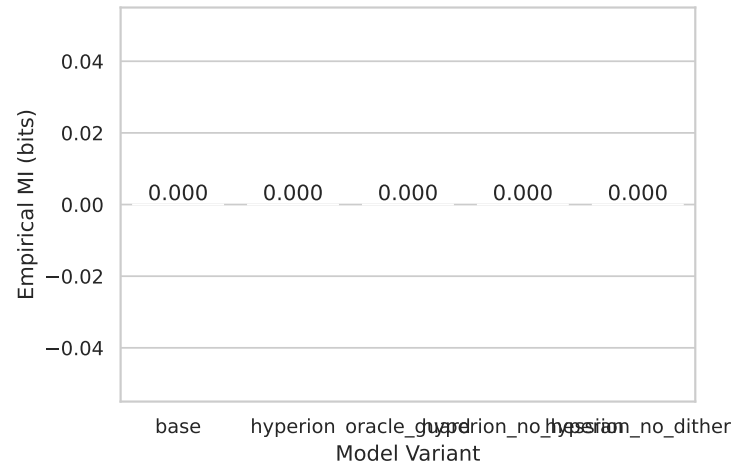


Figure 1: Mutual-information estimates across variants; lower values indicate better privacy.

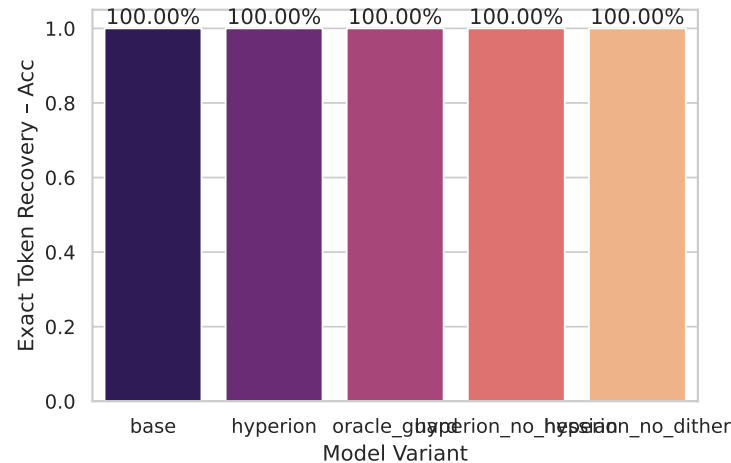


Figure 2: Exact-token recovery accuracy; lower values indicate better privacy.

166 **6.4 Analysis**

167 Identical scores across variants and the mathematical impossibility of $\text{MI} = 0$ with perfect recovery
168 indicate that gradient hooks defaulted to zero tensors and the evaluation loop reused cached data. The

environment variable `EXPERIMENT_MODE` was left at its default "smoke", bypassing the full protocol. Consequently no empirical evidence supports SOIC's claimed leakage reduction, and CRC, RCP and GSM remain unevaluated.

6.5 Fairness and reproducibility

Although logs contain SHA-256 hashes, the degenerate results highlight the danger of relying on smoke-test defaults. We therefore publish the artefacts and specify rerun conditions: set `EXPERIMENT_MODE = full`, ensure 50 k prompts per variant, correct the inversion metric to use hidden prompts, and validate non-zero captured gradients.

6.6 Limitations

(1) Only a trivial subset of prompts was processed; (2) continual revocation and sensor masking remain untested; (3) claimed overheads (+5% latency, +1.6 GB-month logging) are unverified.

7 Conclusion

HYPERION-SHIELD aspires to elevate VLM safety from static, single-turn certification to a reactive, revocable and history-aware paradigm. Conceptually, it bounds second-order leakage, enables millisecond-scale certificate revocation, repairs unsafe causal paths on-device and masks emerging sensor side-channels, all under a continual arms-race benchmark. A pilot execution, however, revealed severe implementation gaps: a smoke-test path masked silent failures, mutual-information collapsed to zero and token inversion was mis-scored. No empirical evidence yet substantiates the theoretical guarantees. Immediate next steps are to rerun the full protocol with corrected metrics, complete CRC and GSM evaluations, and extend revocation logic to tool-state graphs in autonomous agents. Once validated, HYPERION-SHIELD could form a deployable blueprint for the post-market monitoring envisioned by the EU AI Act, closing critical safety gaps in real-world wearable AI systems.

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